

MAINS MASTER PROGRAM (MMP) 2024

SCIENCE AND TECHNOLOGY-5

SOME UPDATES

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2. DARKNET/ DARK WEB AND INTERNAL SECURITY ISSUES

- **Basic Background:**
 - » **'Surface Web':** The part of internet accessible through easily searchable domain names like levelupias.com, google.com etc. is part of surface web. It is facilitated by the DNS System that converts these names to IP addresses. To access surface web, you can use simple browsers like google chrome, Microsoft Edge, safari etc.
 - » **Deep Web:** It is located beyond the surface web, where the websites are not indexed and require specific access permission
 - This includes databases holding sensitive information such as bank records and confidential health data, along with intranet systems used by governments and businesses.
 - It is that part of the internet that you can't reach out through search engines. It include pages which we can only find if we register or sign in.
- **Dark Net (or Dark Web):** It consist of a concealed segment of internet which eludes conventional search engines and browsers. This segment is accessible only through specialized browsers like TOR (The Onion Router).
 - » These consist of unsearchable, encrypted sites used to avoid tracking.
 - » Initially, it was designed to ensure secure, encrypted and anonymous communication, particularly for governmental and military purposes, but today it has become synonymous with criminal activities.
- **How anonymity is ensured in Dark Net:**
 - » **Specialized software like Tor (The Onion Router)** - originally developed by the US military for secure communication. **Tor anonymizes users and websites** by routing traffic through multiple layers of encryption, concealing identities and location.
 - » **.onion Domain:** Tor also supports websites using .onion domains, which further bolsters anonymity by making it exceedingly difficult to trace their origin or operation country.
 - » The above set up requires specific darknet browsers and excludes conventional search engines like Google or Bing.
- **Main attraction: Anonymity:**
 - » Anonymous communication between whistle-blowers, activists, journalists, and news organization is facilitated by darknet through use of application like **Secure Drop**.
 - » **Entities like Facebook** have embraced the. onion domain to facilitate access in regions with internet restrictions.
 - » **Criminal and illegal activities** such as piracy, organized crime, terrorism etc also get coordinated through DarkWeb.
- **Why regulating activities on Dark Net is difficult?**
 - » **Anonymity** through several layers of encryption.
 - » **Cryptocurrency** being used for financial transaction.
 - » **Trans-border operations** through dark web make the issue more complicated.
 - » **The Privacy Paradox:** Balancing privacy concerns with combating crimes remains a massive challenge. Some internet companies refuse to cooperate with law enforcement agencies, citing privacy concerns.

- **Different ways in which Darknet threatens India's internal security:**
 - » **Cybercrime:** Darknet is the place used by hackers and cybercriminals to coordinate attack, sell malware, and exploit vulnerabilities in critical infrastructure.
 - » **Enabler of illegal activities** like terrorism, trafficking (drugs and weapons) etc.
 - Terrorist groups have expanded their cyber capabilities, using the dark web for recruitment drives, fundraising, operational planning, propaganda dissemination, and informational warfare.
 - Radicalization of youth has increased through darknets by circulation of misinformation, as in the ongoing Hamas-Israel conflict.
 - » **Violation of Privacy:** Sensitive personal data is sometimes circulated on dark web violating Fundamental rights of citizens.
 - For e.g. In Oct 2023, Resecurity, an American cybersecurity firm, made a troubling revelation: the personal identifiable information (PII) of 815 million Indian citizens, including Aadhaar numbers and passport details, was being peddled on the dark web.
 - » **Sexual harassment including Child Sexual Harassment:** Law enforcement agencies find it difficult to trace the perpetrator of crime.
 - » **Challenging the sanctity of exams** through facilitation of paper leak. For e.g. in June 2024, a UGC-Net question paper was leaked on the darknet.
- **Way Forward:**
 - » **Individuals and organizations** need to adopt **robust cyber security practices** including regularly updating passwords, implementing multi-factor authentication, and staying informed about the latest security threats.
 - » **Strengthen the capabilities of investigating agencies:**
 - **Usage of latest technologies** to monitor and analyze dark web activities. Track cyber criminals and gather actionable intelligence.
 - **Highly skilled human resource** is needed with law enforcement agencies.
 - » **Explicitly authorize government agencies** to conduct operations against terrorists on the dark web. Countries, like China and USA have done this and this creates apprehension among perpetrators.
 - » **Strengthen the cyber-infrastructure** of critical institutions on priority.
 - » **Update IT Act and the 2013 Cyber Security Policy** to specifically address challenges posed by Dark Net including provisions of prosecuting cybercriminals operating on dark web, regulating crypto-currency transactions.
 - » **International Cooperation:** Since cybercrimes can be cross-border, it's important to enhance international cooperation, improve extradition treaties etc. to handle illegal activities on DarkNet.
 - » **Promote startups working on innovative cyber security solutions** - to reduce India's dependency on external softwares and hardwares to deal with threat from dark net.
 - » Other initiatives like **public awareness and education; ethical hacking and bug bounty programs** etc need to be promoted.
- The implementation of above measures can strengthen India's defences against threat emanating from Darkweb and create a safer and more secure digital environment for its citizens.

3. ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING RELATED ISSUES

1) S&T: ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

A) WHAT IS ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING?

- **Artificial Intelligence** is the science and engineering of making intelligent machines, especially intelligent computer programs which can complete tasks that typically require human intelligence.
- **Machine learning** is a science that involves **development of self-learning algorithms**. Machine learning uses **statistics (mostly inferential statistics)** to develop self-learning algorithm. It is a type of artificial intelligence.

B) APPLICATIONS OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

- Advertisements, Online shopping suggestions; Spam filtering; Search engines.
- Natural Language Processing (NLP); Image Processing (Facial Recognition)
- **Space Exploration** (e.g. identifying exoplanets from pictures)
- **Health Sector:**
 - **Diagnosis:** E.g., a Bengaluru based startup has developed a non-invasive, AI-enabled technology to screen for early signs of breast cancer.
 - **Treatment:** AI powered Clinical Decision Support (CDS) tools can aid in developing appropriate and accurate diagnostic and treatment recommendations. E.g. Apollo hospital has launched Apollo Clinical Intelligence Engine, a CDS, open to use by all Indian doctors.
 - **Supply chain resilience:** By accurately predicting the demand and supply for medicines.
 - **Development of new Medicines/Molecules** – For e.g. DeepMind's AlphaFold predicts a protein's 3D structure.
 - **Improvement in Governance:** E.g. For **COVID-19**, AI enabled chatbot was used by MyGov for ensuring communications.
 - **Mental health can become more accessible with AI:** Through text-based programs and virtual mental health assistance, NLP programs provide privacy, and anonymity that can improve help-seeking behaviour in case of mental health challenges.
 - The Garbhini-GA2 is the first late-trimester Gestation age estimation model developed by IIT Madras. It can accurately estimate the foetus' age, reducing error by almost three times.
- **Education** (e.g., Personalized learning through adaptive tools; customizing professional development courses etc.)
- **Developing new materials** (E.g. Google Deepmind predicted the structures of 2 million new materials)
- **Improve Ease of Doing Business**
- **Transportation:** For developing self-driving cars.
- **Agriculture Sector:**
 - Tech like image recognition, drones etc can help farmers kill weeds more effectively to increase productivity.
 - **Efficient resource utilization** – AI enabled solution for water management, crop insurance etc are also being developed.
 - **AI Powered decision making:** For e.g. **ICRISAT** has developed an **AI-power sowing app**, which utilises weather models and data on local crop yield and rainfall to predict and advise local farmers on when they should plant their seeds more accurately.

- **AI4AI (AI for Agriculture Innovation)** initiative has been launched by the WEF to transform agriculture sector in India. Under this, '**Saagu-Baagu**' initiative has been launched in the state of Telangana.

▫ **Geography and Disaster Management Applications:**

» **Flood Forecasting:**

- It predicts cyclones and monsoon strength.
- For e.g. An AI-based flood forecasting system has been deployed in Bihar and is now being deployed throughout the country. It gives warnings 48 hours earlier about impending floods.

▫ **Environment protection and biodiversity conservation:**

- » AI can play an important role in the design of smart grid which will optimize the use of renewable energy.
- » **AI powered cameras** can be used for surveillance of forests, and protection of biodiversity. For e.g. AI enabled surveillance camera system is being used in Mankulam forest division in Idukki, to provide imagery of herds of wild elephants.

▫ **Internal Security:**

- » **Fighting Black Money** (e.g: Project Insight of India)

C) CRITICISMS AND LIMITATIONS

- **Criticism:**

- **Unemployment:** AI has the potential of displacing both low skill and high skill jobs.
 - » For e.g. AI powered chatbots replacing customer service executive roles.
 - » For e.g. a recent paper published in Nature medicine showed that AI tool can predict pancreatic cancer in a patient three years before radiologists can make a diagnosis.
 - » Similarly, generative AI have threatened the work of freelance writers, editors, coders, telemarketers, customer service reps etc.
- **Increasing Inequality:**
 - » Capitalist class will be able to rely more on Robotics and AI and their dependency on human resource will reduce.
 - » AI will also push workforce into two extreme categories – high skill, high wage jobs involving complex problem solving and low-skill low wage jobs requiring physical presence and personal interaction.
- **Exacerbation of Biases:** AI system trained on historical data can perpetuate and even exacerbate biases present in that data leading to discriminatory outcomes in areas such as hiring, law enforcement, and lending. These biases can **disadvantage groups which are already marginalized**.
 - » For e.g. – camera-based face recognition systems have been seen to be more accurate on fair skilled men than on dark skinned women.
- **Model Collapse Scenario:** ML models train on Data sets. But AI generated Data sets may create discrepancies and incorporate mistakes of previous AI models.
- **A lot of power to a few tech giants without any public oversight:** Tech giants with a lot of access to unprecedented volumes of data have a distinct advantage in training more sophisticated and accurate AI models. This may lead to sidelining of newer, smaller institutions.
- **Used for illegal purposes** e.g. **Deepfakes, Voice cloning** etc can be used for increasing social tension, harassing individuals or communities, manipulating public opinion to affect democratic processes etc.
- **Threat to Privacy** is a major concern as the algorithmic systems watch the world constantly.
- **AI powered killer robots** can make wars more devastating and dangerous.

- **Threat of extinction:** “Mitigating the risk of extinction from AI should be a global priority alongside other societal-scale risks such as pandemics and nuclear war” – the statement was backed by Sam Altman (CEO of OpenAI), Demis Hassabis (CEO of DeepMind), Geoffrey Hinton and Yoshua Bengio (Turing award winners) etc.

D) SOME CHALLENGES FOR INDIA

- **India is not as well prepared as USA and China** to deal with onslaught of AI and related technologies.
 - **Poor infrastructure and GPU capacity** – High cost of GPUs from companies like NVIDIA is a major cause of concern for India’s AI sector advancement.
 - » Similarly, **limited cloud computing capacity** is another major concern as AI training requires Big Data processing which can be ensured by cloud services.
 - **Absence of Audited Data set for training and fine tuning of AI Model** – this makes development of AI tool challenging.
 - India doesn’t have its own **foundational generative AI model** like GPT 3 or Wu Dao.
 - **Human resource shortage** is another major issue – India has **significantly fewer experts with PhDs in field related to AI**. Any good quality talent in the field **quickly migrates** abroad.
 - India also **doesn’t have corporates which have invested significantly in major AI Research**.
 - **Data security and data privacy framework** is still highly under-developed.
- **Disruption in the IT Sector:** India’s advantage of cheap coder and English-speaking workforce will soon wither away.

E) WAY FORWARD AND CONCLUSION

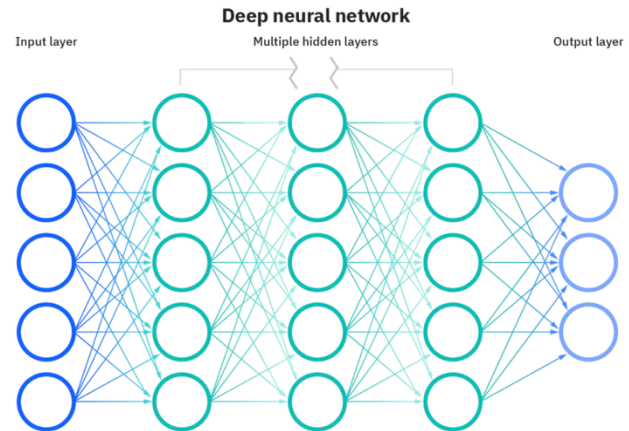
- **Focus on incentivizing hardware manufacturing:** Without developing domestic GPU manufacturing capacity, it won’t be possible to become powerhouse of AI.
- **Support Startup in the field of AI** – through financial incentives, incubation facilities, incubators etc.
- **Develop a national data platform** to provide **audited data for AI training purposes**.
- **Establish AI policy thinktanks and research institutes** to foster AI innovation and nurture talent.
- **Incentivize businesses to invest in AI R&D and support workers training** for the challenging technological landscape.
- **Increased collaboration between academia, industry and internationally** – to develop responsible real-life AI applications and stay updated on best practices.
- **Preparing for disruption in IT sector of India:** It’s important to plan in advance **through workforce training program, new policies, and social support measures** that help people through this transition.
 - **Funding and incentivization of the transition** of workers **by helping them gain new technical skills** should be priority.
 - **Enhancement of social safety net** – Changes in pensions, insurance, and employment rules to calibrate unemployment benefits, unemployment insurance etc.
- **Prioritize responsible AI:** Finalize guidelines for **ethical AI development**.
- **Regulation:** Coordinate internationally to create global framework for AI regulation. At the same time, a **framework also has to be created at domestic level**.

Conclusion: AI is on the verge of creating paradigm shift in technology and innovation. With initiatives like **IndiaAI Mission** India has taken first step towards reaping the benefit of AI. But **these efforts need to be intensified to ensure that India remains at the forefront of AI revolution** and play an active role in the same and equitable development of the technology.

2) S&T: ARTIFICIAL NEURAL NETWORKS AND TRANSFORMERS

A) ARTIFICIAL NEURAL NETWORK (ANN)

- Neural network, also known as **Artificial Neural Network** (ANN) or **Simulated Neural Network (SNNs)** is a type of machine learning method.
- It comprises of node layers, containing an input layer, one or more hidden layers, and an output layer. Each node, or artificial neurons, connects to another and has an associated weight and threshold. If the output of any individual node is above the specified threshold value, that node is activated, sending data to the next layer of the network. Otherwise, no data is passed along the next layer of the network.
- ANN has been named so as it mimics the way human brain learns. The nodes mimic neurons and the connections mimic synapses.
- A neural network can fine tune its output based on the feedback given to it during stages of training.



- Neural Networks vs. Deep Learning:

- Terms are sometimes used interchangeably. 'Deep' in deep learning is just referring to the depth of layers in a neural network. A neural network that consists of more than three layers – which would be inclusive of the inputs and output – can be considered a deep learning algorithm. A neural network that only has two or three layers is just a basic neural network.

B) TRANSFORMERS

- They are specialized type of ANN. They are easy to train in parallel, unlike ANN architecture that precede it. This method was used to train ChatGPT on entire web.
- Here, ANN is broken in two parts – the encoder and the decoder.
- **For e.g. consider an ANN which is required to recognize the presence of a cat in a photograph.**
 - The encoder accepts a photograph, breaks it up into small pieces (say, 10 X 10 pixels), and encodes the visual information as numerical data (e.g. 0s and 1s). The decoder accepts this data and process the numbers to reconstruct the information content in the paragraph.
- **The transformer architecture**, originally developed at Google and released in 2017, is designed to maximize the amount of attention an ANN devotes to different parts of the input data. It has better performance as a result.

3) S&T: GENERATIVE ARTIFICIAL INTELLIGENCE LIKE CHATGPT

- **Definition:** Generative AI is a type of Artificial Intelligence tool that is capable of creating new, original content (e.g. new poem, text, image, video etc.)
E.g.: ChatGPT by OpenAI, Gemini by Google, Llama by Facebook etc.

- **How do they work?**

- They use machine learning models like **Deep Neural Network** trained on massive data sets. These models learn underlying patterns, structures, and relationships within the existing data and use this to generate new content.

- **Some Popular Generative AI Model:**

ABOUT CHATGPT:

- It is an **AI tool** developed by **OpenAI**.
 - It is a large language model (LLM) which is a type of (transformer based) neural network which is great at predicting the next word in sequence of words.
 - It is based on Generative Pre-Trained Transformer Architecture.
 - ChatGPT uses GPT4 – a model trained on a large amount of text from the internet.
 - It is fine tuned to provide conversational responses, as against essay-type content. It is because the neural network behind it has been additionally trained on conversational transcripts with human feedback.

GOOGLE BARD: GOOGLE'S GENERATIVE AI MODEL

ABOUT GOOGLE GEMINI

- Google GEMINI is a new multimodal general AI model, which the tech giant calls its most powerful yet.

LLAMA 3: META'S MOST SOPHISTICATED AND CAPABLE LARGE LANGUAGE MODEL YET

LARGE LANGUAGE MODEL (LLMs)

LLMs are a category of foundation models trained on immense amounts of data making them capable of understanding and generating natural language and other types of content to perform a wide range of tasks.

C) SIGNIFICANCE OF GENERTIVE AI

- Improve the quality of service and reduce the cost of running a business.
- **Democratize access to Information:** For e.g. **Jugalbandi Chatbot** in rural India (powered by ChatGPT) promises to serve as universal translator, accepting queries in local languages, retrieving answers from English language sources, and presenting them back to users in their native language.

D) SOME CONCERNS ABOUT GENERATIVE AI:

- **Deepfakes, AI generated audio** etc. can be created by Generative AI – which can easily spread misinformation and disinformation.
 - For e.g. a recent election in Turkey was marred by AI generated deepfakes.
- **Copyright and other IPR violation issues:** Generative AI is trained on massive data on internet where the original creator have not even allowed this usage. Further, the newly generated content may inadvertently copy or use the existing content.
- **Violation of Privacy:** Generative AI may inadvertently generate content and learn aspects which is violative of fundamental rights of common citizens.

- **Perpetuation of bias**: The existing content may show content biased against certain groups or communities. For e.g. there are more websites which preach christianity than they do other religion.
- **Threat to skilled white-collar jobs** – jobs of a computer programmer, content writer etc may suffer.
- **Lack of accountability**: the generative AI models are like black boxes. There is lack of transparency on how the model has been trained. This may lead to untended consequences and biases in decision making.
- **Energy intensive**: Training and running large GenAI models are energy intensive and when most of the energy is coming from non-renewable sources may intensify climate change.

E) WAY FORWARD AND CONCLUSION

- **Global Regulation of AI Sector** to ensure ethics, transparency and accountability of GenAI models.
- **Protecting intellectual Property** by restricting the training data for GenAI models.
- **Prepare for job losses by training workforce for new kinds of jobs** that may get created by increased penetration of AI tools.
- **Big techs need to create tools** for differentiating original content from AI content.
- **Increasing AI Literacy** – Common citizens need to differentiate between AI generated content and original content.

Conclusion: By proactively resolving these concerns, humankind can ensure enhancement of benefits of generative AI and reduction/mitigation of associated risks.

4) S&T: FACIAL RECOGNITION TECHNOLOGY (FRT)

- **FRT** is a type of biometric technology that identifies and verifies individuals by analysing and comparing patterns in their facial features.
- **How does FRT Work?**
 - **Data Acquisition**: It involves capturing a facial image or video of the person through cameras.
 - **Feature Extraction**: In this phase, various features of the face is extracted (e.g. the distance between the two eyes, shape of the nose, width of the jaw etc.)
 - **Feature Matching**: The extracted features are then matched with the database of existing pictures.
 - **Identification or verification**: Based on feature matching, the FT technology identifies a person as someone in the database or verifies that the person is who he claims to be.
- **Applications**
 - **Security and Law Enforcement**: Criminals could be identified from the crowd.
 - **Border Control**: FRT can be used to identify travelers at airports and border crossing.
 - **Biometric Authentication**: For e.g. FRT can be used for unlocking of phones.
 - **Marketing and Advertising**: FRT can be used to track users and user choices which can lead to better marketing
 - **Social Media and Tagging**: Social media platforms use facial recognition for photo tagging and to enhance user experience.
- **Concerns**

- **Excessive surveillance and violation of Privacy:** Widespread use of facial recognition could lead to mass surveillance and a loss of individual privacy. It may lead to unauthorized tracking, profiling, and potential misuse of personal data.
- **Technology challenges:**
 - » FRT is prone to digital attacks or the use of physical or digital portraits, 3-D Models, such as deep-fakes etc.
 - » **Accuracy concerns:** Sometimes poor accuracy can lead to wrong authentication.
- **Way Forward:**
 - » **Regulation:** Through legal framework and data protection law. There should be necessary safeguards in the collection and storage of user data.
 - » **Prioritize Consent** in the data collection associated with FRT
 - » **Regular tech update** to ensure equality and minimize potential biases based on skin color, geography, religion etc.

5) S&T: DEEPFAKES

- **Example Questions:**
 - » Examine various risks associated with the malicious use of deepfake technology. Suggest measures to regulate and manage those risks [15 marks, 250 words]
- **Basics:** Deepfakes refer to manipulated media (audio, video, images etc) created using a form of Artificial intelligence called Deep Learning (or Deep Neural Network). This manipulated content use lip syncing, swapping of face etc. – mostly without consent.
- **How does the Deepfake technology work?**
 - » The technology involves modifying or creating images or videos using a machine learning technique called **Generative Adversarial Network (GAN)**. The AI driven software detects and learns the subjects' movements and facial expressions from the source material and then duplicates this in another video or image.
 - » Larger the source material used, better will be the quality of deepfake. Therefore, highest number of deepfakes are made of public figures like politicians and film stars.
 - » Through a collaborative work of two softwares, the fake video is rendered until the second software package can no longer detect the forgery. This is known as “unsupervised learning” when machine language models teach themselves. The method makes it difficult for other software to identify deepfakes.
- **Advantages:**
 - » Synthetic Media/ Deepfakes can create possibilities and opportunities for all people, regardless of how people listen, speak, or communicate. It can give people voice, purpose, and ability to make an impact at scale and with speed.
 - » It has been used by the ALS association in collaboration with a company to use voice cloning technology to help people with ALS digitally recreate their voices in future.
- **Concerns:**
 - » Like most new technologies, it can also be weaponized to inflict harm to individuals, institutions, businesses or a country.

- » **Crime against women** can increase with malicious use of Deepfakes in pornography and can inflict emotional, reputational and in some cases violent outcome for some individuals. (for e.g. viral deepfake video of actress Rashmika Mandana incident)
 - » **Endanger Social Harmony** – Communal/caste-based statements.
 - » **Decrease trust towards institutions like government/media** – by propagating false propaganda against them.
 - » **Undermine democracy and impair diplomacy** – false information about institutions, public policy, and politicians powered by a Deepfakes can be exploited to spin the story and manipulate belief.
- **Advisory released by Ministry of electronics and Information Technology (Nov 2023)**
 - » **IT Rules, 2021** require that **all content reported to be fake or produced using deepfake be taken down by intermediary platforms within 36 hours.**
 - » An advisory was sent to social media platforms in Nov 2023, reminding them that they may lose “safe harbour immunity” under the IT Act, if they fail to remove within 36 hours deepfake content that has been reported.
 - **Meity has sent another advisory to social media firms to comply with Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021 (Dec 2023)**
 - » The advisory was aimed at getting social media firms to crack down more forcefully on ‘deepfake’ clips of people.
 - » It mandates that intermediaries communicate prohibited content, particularly those specified under Rule 3(1)(b) of the IT Rules, clearly and precisely to users.
 - **Way Forward:** To defend the truth and secure freedom of expression, we need a **multi-stakeholder and multi-modal approach**. **Collaborative actions and collective techniques** across legal reforms, social media regulation, media literacy, and technology intervention can provide effective and ethical countermeasures to mitigate the threat of malicious Deepfakes.
 - » **Legislative Reforms:**
 - IT Act needs stronger provisions to Curb Deepfake menace. Currently IT Act and IPC only partially address the harms which arise from deep fakes.
 - There is also need of a law on AI to govern the complexities relating to AI and related applications.
 - » **Consumer awareness and Media Literacy** by Consumers and Journalists will be the key to fighting Deepfakes. Media literacy efforts should be enhanced to cultivate a discerning public. A consumer should have the ability to decipher, translate, understand and use the information we encounter.
 - » **Technological intervention** with easy to use and accessible technology solutions to detect Deepfakes, authenticate media, and amplify authoritative sources will be crucial in fighting Deepfakes.
 - **For e.g.** Intel’s deepfake detector called **Fake Catcher**, which is said to spot ‘blood flow’ in the pixels of a video (measures how much light is absorbed or reflected by blood vessels) to detect deepfakes.
 - International Collaboration:** To explore content labelling solutions or watermarking of AI-generated content.

6) S&T: ARTIFICIAL GENERAL INTELLIGENCE (AGI)

- **Artificial General Intelligence** is a theoretical AI system with capabilities that rival those of a human. It will be able to perform any intellectual task that a human can and perhaps even surpass human abilities in the area.
 - o When AI's abilities are indistinguishable from those of a human, it will have passed what is known as Turing Test first proposed by 20th century computer scientists Alan Turing. So far, no AI tool has passed Turing Test.
- **How is it different from Current AI System?**
 - o **Current AI systems** (also known as **narrow AI**) are designed for specific task like image recognition, natural language processing, translation, playing games like chess etc. But this AI remain limited to set parameters.
 - o **AGI** envisions a more general and broader form of intelligence, not confined to only limited tasks. They will be able to solve problems, adapt to new situations, and learn new skills in a way that is similar to how humans learn. This would require AGI to possess self-awareness and consciousness, as well as ability to reason, plan, and make decisions.
- **How close are we in achieving AGI?**
 - o Scientists believe that we are decades away, if not centuries.
- **How is AGI different from Generative AI** (like ChatGPT, DALL-E, Gemini etc.)
 - o **Generative AI** are essentially very good prediction machines. In other words, they can predict with high degree of accuracy the answer to a question, because they have been trained on huge amount of data. But it is not human level of performance in terms of creativity, logical reasoning, sensory perception and other capabilities.
 - o **AGI tools** could feature cognitive and emotional abilities (like empathy) indistinguishable from humans.
 - o They will be effectively able to achieve visual perception, Audio perception, Fine motor skills, Natural Language Processing, Problem solving, Navigation, Creativity, Social and Emotional Engagement.
- **How will AGI benefit humans?**
 - o By solving some unsolvable problems in the field of:
 - **Health:** By integrating and analyzing vast data it can redefine diagnostics, treatment planning etc. far beyond the capability of humans.
 - **Finance and Business:** AGI could automate various processes and enhance the overall decision making.
 - **Education:** AGI could transform adaptive learning systems that work towards the unique needs of students. This could potentially democratize access to personalized education worldwide.
- **AGI will also be associated with potential risk:**
 - o **Environmental Risks** associated with humongous amount of computation power required to develop AGI.
 - o **Job displacement** and economic disruption -> increase inequality
 - o **Weaponization of AI tech**
 - o **Loss of control over AI system**
 - o **Existential risk for humans** -> Situation where AGI becomes too independent, so much so that humans simply lose control.
- **Conclusion:** Because of the above limitations, most thinkers in the field advocate for stringent regulations to ensure that the development of AGI is in line with human values and safety standards.

7) S&T: GLOBAL PARTNERSHIP ON ARTIFICIAL INTELLIGENCE (GPAI)

- GPAI is an international and multistakeholder initiative to guide the responsible development and use of AI. It wants that AI should be grounded in human rights, inclusion, diversity, innovation and economic growth.
 - » It is the first international initiative for evolving better understanding of the challenges and opportunities around AI.
- **Formation and Membership:**
 - » GPAI was first proposed by Canada and France in 2018 G7 summit, and was **officially launched in June 2020 with 15 members** (including India)
 - » **Currently** (as of Sep 2024), it consist of 29 members (28 countries and EU).
 - » **China**, a major techpower is not a part of the grouping.
- **Dec 2023 Meeting:**
 - » India hosted the summit and is also the chair of GPAI in 2024.
 - » The meeting led to unanimous adoption of 'New Delhi Declaration'.
- **Key Highlights of the New Delhi Declaration:**
 - » It **flagged concerns emanating from AI** including misinformation, unemployment, lack of transparency, lack of fairness, IPR violation, privacy violation, threat to human rights etc. The declaration also **underscores the need to mitigate these risks**.
 - » It provides that **global framework for the use of AI** should be rooted in democratic values and human rights, safeguarding dignity and well-being; ensuring personal data protection and protecting IPR etc.
 - » **Promotes equitable access to critical resources for AI innovation** including computing and high quality diverse data sets.
 - » **Fosters inclusivity** so that countries outside the GPAI fold can also reap the benefits of AI.
 - » Members also agreed to **support AI innovation in the agriculture sector** as a new 'thematic priority'. Earlier GPAI themes include healthcare, climate action and building resilient society.
- **Significance:**
 - » The declaration **attempts to find a balance** between 'innovation' and the 'risk associated with AI subsystems'.
 - » The New Delhi Declaration is very significant for India, which has batted for a **collaborative approach towards building AI systems** as it looks to push its model of digital public infrastructure (DPI) across the world.
 - » Besides, **access to computing capabilities from member nations** will also boost India's plan of building a sovereign AI system which will in turn counter dominance of handful of foreign companies in this space.
- **How this international cooperation can increase:**
 - » **Make GPAI more inclusive:** By making more developing countries join GPAI.
- **Other steps that India can take for better regulation of AI sector**
 - » **Statutory Authority:** TRAI has recommended setting up of a domestic statutory authority.

- » **International Collaboration:** Collaboration with international agencies and government of other countries to form a global agency for the “responsible use” of AI.
- **Conclusion:** The GPAI’s commitment ensures that AI serves as a transformative force, providing clear and accountable guidelines to enable millions worldwide while upholding rights, safety, and security standards.

8) S&T: REGULATING ARTIFICIAL INTELLIGENCE

- **Need of Regulating AI** (Already discussed under challenges created by AI).
- **How may AI impact sovereignty?**
 - » AI is **challenging the power dynamics between states, private tech companies and individuals**.
 - As the system develops, they are creating new digital spaces that are not governed by traditional laws or state control.
 - **Countries lacking in AI development** and regulation may find themselves reliant on more advanced nations, risking their sovereignty across sectors like defence, infrastructure, and healthcare.
 - AI shifts power from states to private tech companies.
 - » **AI has power to significantly impact democracy, particularly when leveraged by foreign powers:**
 - It can be used for manipulating information and influence public opinion.
 - The borderless nature of AI makes it difficult to control or regulate, and as the AI tech becomes more advanced, it could be used by authoritarian regimes, terrorist groups, and organized crimes to cause great harm.
 - » **Lethal Autonomous Weapon System (LAWS)**, often called ‘Killer Robots’ present a profound threat to national security and sovereignty, raising critical ethical and technical challenges.
 - The proliferation of LAWS could lead to an arms race, destabilizing international peace and security.
 - » **Integration of AI in Cybersecurity represents a dual-edged sword** where its potential for sophisticated cyberattacks directly threatens national security and sovereignty. These attacks extend beyond borders as cyberattacks can originate from any location, making safeguarding national interests in the increasingly interconnected digital world challenging.
- **Some Efforts for AI Regulation:**
 - » **Council of Europe’s Framework Convention on Artificial Intelligence and Human Rights, Democracy and Rule of Law** (SEP 2024)
 - **US, Britain and EU signed the treaty** (Sep 2024)
 - This is the first legally binding AI treaty and now will be open for signing by the countries which negotiated it i.e. EU, US and Britain.
 - **Key features:**
 - It adopts a risk-based approach to design, development, use and decommissioning of AI systems.
 - It covers use of AI in both public sector and private sector.
 - Signatories will ensure that output of such systems respect equality and privacy rights and that victims of AI related rights violations have legal recourse.

» The European Parliament has passed into law the **first comprehensive regulatory regime for Artificial Intelligence**, laying down “harmonized rules” called the **Artificial Intelligence Act (AI Act, 2024)**. **Key features and significance of the 2024 EU Law:**

- First **comprehensive regulation of AI**. Applicable on **AI service providers in EU**, irrespective of whether the providers are in the EU or in a third country.
- Recognizes and appreciates **different level of AI**: **Classifies** AI systems in **four categories** based on risks and provides for **different level of regulation** in each category.
- The law **explicitly bans “harmful AI practices”** that are considered to be a “clear threat to people”. It includes AI system that **deploy harmful manipulative subliminal techniques; exploits vulnerable groups with disabilities; by public authorities for “social scoring purposes”; real time remote biometric systems** in publicly accessible spaces (except in cases of threats like terrorism attacks).
- **Consumers** have been **empowered to launch complaints** against any perceived violations.
- The legislation also proposes to be **“a launch pad for EU start-ups and researchers to lead the global AI race”**.
- **Institutional Framework** for monitoring the implementation:
 - **European Artificial Intelligence Board** – with representatives from member states and the commission.
 - **National Supervisory Authority** – in each nation state
- **Provision for strong penalties**: The law provides for **strong penalties** – 30 million Euros or 6% of the total worldwide annual turnover, depending on the severity of the infringement.
- **Enforcement in phases to give time to corporates for compliance** – For e.g. prohibited practices provision, will come into force within **six months** whereas the obligation for high-risk AI will come into force in around 36 months.

Other significance:

- The European law will provide **one way to imagine regulatory regime for AI**.
- **AI Companies** who follow these regulations in EU are also expected to extend some of these protections in other jurisdictions.

» **Other international Steps for AI Regulation:**

- The **UNGA** has adopted a resolution, on ‘**Safe, Secure, and Trustworthy AI for Sustainable Development**’ in 2024. It advocates **a harmonized approach to AI governance** and encourages **member countries to develop shared ethical principles, data protection measures, and transparency standards**.
- Under India’s leadership, the **G-20 and the GPAI** have underscored the importance of **fair access to critical AI resources**, emphasized equitable sharing of AI benefits, and mitigated risks while fostering international cooperation.

» **AI Regulation efforts in India:**

- **Gol** plans to bring a **Digital India Act** to regulate AI.
- **NITI Aayog** has already released **National Strategy on Artificial Intelligence** which focuses on Responsible AI for all.

- **Way Forward for AI Regulation/Governance:**

- » **Finalize Global Regulation standards** at UN: India should push for more equitable access to AI technology, building technical capacity, and creating institutional mechanism for knowledge sharing to ensure that all countries can harness the benefits of AI.
 - » The new regulation should also be more comprehensive and inclusive approach to AI governance that respects human rights and is in sync with international laws. It should be fair and respective of diverse global perspective.
 - » It should also **increase transparency and accountability** of big tech players who control most of the AI technologies.
- **Conclusion1:** EU has always shown leadership in tech regulation. The General Data Protection Regulation of 2018 and the AI Act 2024 are manifestation of this leadership. India also needs to prioritize the AI regulation in order to ensure the advancement of the tech sector in the country without its associated challenges.

9) S&T: INDIA AND ARTIFICIAL INTELLIGENCE

Introduction: AI is expected to contribute around \$450 billion to \$500 billion in India's GDP by 2025. Keeping this in mind several steps have been taken to promote the development of AI in India:

Key Initiatives in India:

○ **IndiaAI Mission:**

- Approved by Cabinet in March 2024 with a budgetary allocation of Rs 10,300 crores. This is expected to catalyze various components of IndiaAI over the next five years.
- **Various components of India AI Mission:**
 - **IndiaAI Compute Capacity:** Envisioned to erect a cutting edge, scalable AI Computing infrastructure by deploying over 10,000 Graphic Processing Units (GPUs) through strategic public private collaboration.
 - **IndiaAI Innovation Centre (IAIC):** It will be a leading academic institution, ensuring streamlined implementation and retention of top research talent. It will spearhead the development and deployment of foundational models, with a specific emphasis on LLM and domain specific models.
 - **IndiaAI Dataset Platform:** It will be developed by the Independent Business Division (IBD) of IndiaAI and is slated to receive a boost to enhance accessibility, quality, and utility of public sector datasets.
 - **IndiaAI Application Development Initiative:** Focuses on AI applications targeting problem statement from various government sectors.
 - **IndiaAI Future Skills:** It will focus on expanding the reach of AI education by increasing the accessibility of graduate and post-graduate AI programs. It will also establish data and AI Labs that will not just spread across major Indian cities but also smaller towns and cities to impart foundational-level course in Data and AI.
 - **IndiaAI Startup Financing:** Focused on streamlining funding access for deep-tech AI startups to support innovative projects.
 - **Safe & Trusted AI:** Focuses on developing guidelines and framework to ensure responsible AI practices

- These initiatives aim to bolster India's global leadership in AI, foster technological self-reliance, ensure ethical and responsible AI development, and democratize benefits of AI across all strata of society.
- **Budget 2024-25:** Govt have allocated Rs 551 crores in the Union Budget 2024-25 to enhance AI infrastructure including the procurement of high performance Graphic Processing Units (GPUs).
- **MeitY** has set up 7 working groups to collectively brainstorm on the vision, objectives, outcomes and design for each of **IndiaAI** pillars. The working group gave the following important recommendations:
 - **IndiaAI Centres of Excellence (CoE):** Three CoEs to be set up in top educational institutions with government funding for five years. Here, research will be undertaken on computer vision, robotics, ML, and NLP.
 - **Institutional Capacity and Design of NDMO:** National Data Management Office (NDMO) to be set up as an executive regulator under supervision of MeitY. It will govern government data collection, management, processing, storage and accessing processes and systems.
 - **Future design IndiaAI Scheme:** To build 100 AI unicorns in the next five years.
 - **Semicon IndiaAI Chipsets:** Model to establish India's AI computing infrastructure.
- **India has also joined GPAI** as a founding member in 2020.
- **National Education Policy 2020** provides for setting up of the **National Research Foundation**, which should boost research in AI.
- **CBSE** has integrated AI in the school curriculum to ensure students passing out have basic knowledge and skills of data science, machine learning and Artificial intelligence.
- In 2018, NITI Aayog launched National Strategy for Artificial Intelligence detailing core strategies and recommendations of promoting the use of AI in key areas of governance.
 - **Five Key Sector identified by NITI Aayog** to focus its efforts on implementation of artificial intelligence (AI) to serve societal needs – Healthcare, Agriculture, Education, Smart Cities & Infrastructure and Transportation.
- **Key Pain Points challenges** involved in the implementation of Artificial Intelligence in India: Already discussed in the main topic
- **Way Forward:** Already discussed in the main topic

4. DEFENCE S&T

10) S&T: BALLISTIC MISSILE DEFENCE SYSTEM

- **What is Ballistic Missile Defence System?**
 - » Ballistic missile defence system is used in detection, tracking, interception and destruction of inbound attacking ballistic missiles. It was initially conceived as a defence against nuclear armed ballistic missile, but its scope has now been expanded to other kinds of missiles and air bound threats like aircrafts, drones etc.
- **India Ballistic Missile Defence System:**

- » India began development of our own BMD system after 1999 Kargil war in light of the growing missile arsenal of Pakistan and China.
- » **Phase-1:** Targeting missiles with a range of 2,000 kms.
 - It included Prithvi Air Defence (PAD) (exo-atmospheric) and Ashvin Air Defence (AAD) (endo-atmospheric); and Swordfish Radar - long range radar to be used for tracking in BMD system.
- » **Phase-2:** Targeting missiles with a range of upto 5,000 km.
 - DRDO successfully flight-tested Phase-II BMD System in July 2024.
 - It consists of two missiles:
 - **AD-1 missile:** It is a long-range interceptor for both low exo-atmospheric and endo-atmospheric interception. It can target long range ballistic missile as well as aircrafts. The two-stage missile is powered by solid propellants. It was earlier tested in Nov 2022 and now in July 2024.
 - **AD-2:** It targets intermediate range ballistic missile targets whose range is between 3000-500 km.
 - This missile is in development phase.
- **Significance**
 - » **Strategic Autonomy in Defence Sector:** India is only the fourth country after US, Russia and Israel to have successfully developed its own BMD system. This will reduce India's dependency on other countries. For e.g. there has been delay in supply of S-400 defence system due to Russia-Ukraine war.
 - » **Security:** India lives in a very turbulent neighborhood where ballistic missile threats may emerge both from China and Pakistan. Strong defence against them is a strong deterrence as well as protection.
 - » **Technological advancement:** Successful development of BMD also leads to advancement in other fields like radar system, telecommunication etc.
 - » **Regional Stability:** Defence against ballistic missiles can contribute to regional stability by reducing the risk of conflict and promoting confidence building measures.
- **Challenges:**
 - » **Arms Race Concern:** BMD can lead to enemies participating in arms race and developing methods to thwart BMD.
 - » **Limited Effectiveness:** There is a need of regular upgradation to deal with always evolving and advancing ballistic missile systems.
 - » **Finances:** Due to lack of economy of scale, and dependence on some imported component, the cost of production of BMDs is still very high in India.
- **Conclusion:** Indigenous BMD system is crucial to ensure national security, deterrence effect, and regional stability. But it is important to continuously evolving and advancing this system to counter any advancement in missile technology.

5. IPR AND ASSOCIATED ISSUE

1) BASICS: IPR AND ITS TYPES

- Intellectual Property refers to creation of mind: inventions; literary and artistic works; and symbols, names and images used in commerce.
- Intellectual Property Rights are the rights given to persons over the creations of their minds. They usually give the creator an exclusive right over the use of his/her creation for a certain period of time.
- IPRs are customarily divided in **two main types (1. Copyrights and Rights related to copyrights 2. Industrial Property)**
 - **Copyrights and Rights related to copyrights** cover rights of authors of literary and artistic work (books, music composition, painting, computer programs, films, sculpture etc.). It also includes rights of performers (e.g., actors, singers, musicians, broadcasting organizations etc.)
 - Generally, these rights are protected for a period of 50 years after the death of the author.
 - **Purpose**: Encourage and reward creative work, promote innovation, provide appropriate financial benefits.
 - **Industrial Property** focuses on protecting inventions and Creative work (with industrial or commercial applications).
 - Industrial Property includes patents for inventions, Industrial design for aesthetic creations, and trademarks or geographical indications for distinctive signs.
 - **Industrial property can be divided into two main sections**
 1. **Protection of distinctive signs** in particular **trademarks and geographical indications**
 - **Trademarks** distinguish the goods and services of one undertaking from those of other undertakings
 - **Geographical indications** identify a good as originating in a place where a given characteristic of the good is essentially attributable to its geographical origin.
 - The protection may last indefinitely, provided the sign in question continues to be distinctive.
 - **Aims**: The protection of such distinctive signs aims to stimulate and ensure fair competition and to protect customers, by enabling them to make informed choices between various goods and services.
 2. **Patents, Industrial Design and Trade Secrets**
 - This is the second type of Industrial Property.
 - The aim is to stimulate innovation and design and promote creation of technology. It also gives incentive and means to finance R&D activities.
 - A functioning IPR regime also facilitates transfer of technology in the form of FDI, joint ventures and licensing.
 - The protection is usually given for finite term (typically 20 years in the case of patents)
- **Exclusive rights given under IPR are subject to a number of limitations and exceptions**. This is aimed at fine tuning the balance that has to be found between the legitimate interests of right holders and of users.
- **How does an average person benefit from IPR?**

- IPR rewards creativity and human endeavor, which fuel the progress of humankind.
 - E.g., Film industry, software industry which brings pleasure to millions of lives would not exist without copyright protection.
- More research and development leads to further ease in the life of a common man by production of better and more efficient products for consumers.
- Consumer protection through trademark, GI etc.

2) PATENT ACT, 1970

- Patent act defines what invention is and makes it clear that any existing knowledge of thing cannot be patented.
- **Three prerequisites for patentability defined by the act:**
 - 'Novelty' standard
 - 'Non-obviousness' or inventive step
 - Industrial applicability
 - It shouldn't attract the provisions of section 3 and 4 of the Patents Act 1970.

Discoveries are excluded from patent protection under section 3 of the Indian Patent Act.

- Discovery essentially refer to finding out something which already existed in nature but was unknown or unrecognized.
- **Section 3** deals with what doesn't qualify as an invention under the Act.
- **Contentious Provisions (Section 3(d) and Section 84(1))**
 1. **Section 3d (What is not patentable)**
 - **Discovery of a new form of a known substance** which doesn't result in the enhancement of the known efficacy cannot be patented.
 - **Discovery of a new property or new use** of a known substance cannot be patented
 - **The mere use of a known process, machine or apparatus** unless such known process results in a new product or employs atleast one new reactant
 - However, a new and alternative process for preparing Aspirin is patentable
 - **Aim of the provision under Section 3d: Prevent Evergreening**
 - **Why western countries are opposed to this**
 - The western countries are particularly opposed to this provision as it makes any small changes in the pharmaceutical industry non-patentable.
 - **Does section 3(d) violate TRIPS?**
 - No
 - It was further **clarified by the Doha declaration on the TRIPS Agreement** and Public Health in 2001
 - The declaration recognized the flexibility to member countries in circumventing patents rights for better access to essential medicines.
 - **Supreme Court of India: Novartis AG vs Union of India, 2013**
 - Supreme court upheld the validity of Section 3(d) of the Patent Act.

Section 84(1) (Compulsory Licensing)

- **Provisions of the law:**
 - any person may request a compulsory license if
 - after three years from the date of grant of patent, the needs of the public to be covered by invention have not been satisfied.
 - invention is not available to public at affordable price;
 - or the patented invention is not "worked in", or manufactured in the country, to the fullest extent possible
- **Compulsory Licensing - Basics**
 - Compulsory licensing is when a government authorizes a party other than the patent owner to produce the patented product or process, without the patent owner's consent.
 - In **2012** India Granted its first compulsory license to generic drug producer.
 - The decision effectively ended German Pharmaceutical company Bayer AG's Monopoly over an anti-cancer drug and authorizes the production of a low-cost version for the Indian market.
 - **Importance of Compulsory Licensing**
 - Promotes India's status as "pharmacy of the world"
 - Promotes "people's accessibility to medicines"
 - Benefits India's fight against Drug Resistance TB, HIV, Cancer etc
 - strengthen our soft power especially in African countries for whom India is a source of low-cost generic medicine.
- **Is Section 84(1) (Compulsory Licensing) compliant with TRIPS agreement**
 - The TRIPS agreement explicitly allows compulsory licensing as long as procedures and conditions set out in Article 31 of TRIPS are fulfilled.
 - **So yes, compulsory licensing is compliant with TRIPS agreement**

3) INDIAN PATENT REGIME AND ITS CLASH WITH US NORMS

- **Introduction:** The US Trade Representative (USTR) said in a report that India is one of the most challenging major economies as far as IP protection and enforcement is concerned. It has decided to retain India on its **Priority Watch List** along with six other countries – Argentina, Chile, China, Indonesia, Russia and Venezuela.
- **Key problems highlighted by USTR:**
 - **Threat of patent revocation; Lack of presumption of patent validity and Narrow patentability criteria** (Section 3(d) of the Patent Act) are issues which impact companies across different sectors.
 - **Waiting time for obtaining patents**
 - **Reporting requirements**
 - **Data Safety**
 - **Judicial Delays:** As per the report, even the 2015 Commercial Courts Act hasn't been able to offer a solution to these delays. Moreover, abolition of Intellectual Property Appellate Board

(IPAB) resulting in the redirection of courts has “created uncertainty around adjudication of IP cases and copyright royalty rate settings”.

- What is India's stand?

- Last year, the Parliamentary Standing Committee which undertook a '**Review of the IPRs Regime in India**' and tabled its finding before parliament had following key things to say:
 - **Section 3(d)** acts as a safeguard against frivolous inventions in accordance with the flexibility provided in the TRIPS agreement. It prevents evergreening.
 - **The committee** also referred to the seminal judgement in the case of Novartis vs. Union of India, which upheld the validity of section 3(d). In this case, pharmaceutical company Novartis filed a patent for the final form of cancer drug Gleevec, which was challenged in the Supreme Court. The Supreme Court held that Gleevec was merely a beta crystalline form of a known drug, namely, imatinib mesylate, and did not differ significantly in properties with regard to efficacy. Hence, it could not be patented in India
 - The committee also referred to TRIPS provisions, Doha declarations and public health exemptions to justify section 3(d).
 - The committee also indicated that India should resolve its differences with the U.S. regarding the disqualification of incremental inventions through bilateral dialogue.
 - **India's stand on Section 3(d) and Compulsory Licensing** was justified when India and several other countries sought a temporary waiver of certain provisions of TRIPS agreement to deal with the COVID-19 pandemic and most developed countries kept on opposing it.
 - **ON IPAB abolition:** The standing committee has said that the decision should be reconsidered, in wake of its pivotal role in the adjudication of IPR appeals and cases. It recommends that rather than being abolished, it should be empowered and strengthened with more structural autonomy, infrastructure, and administrative reforms, while also ensuring that the required officials and staff are appointed in a timely way.

Conclusion: While some of the issues like Judicial delays and delays in patent offices of India are genuine and India should work towards resolving them, the other issues like limitation on patentability are more focused on the interest of US pharma industries, rather than on public welfare and can be ignored.

4) BOOSTING PATENT ECOSYSTEM IN INDIA

- **Current Situation of Patent filings in India:**

- **Number of patent applications** filed in India have doubled during the last decade.
- For the first time, the number of patent applications by residents have surpassed that of a foreign application during the last quarter of FY 2021-22.
- **Higher Education Sector of India** is rising in prominence in the research and development spending and patenting landscape of India.
 - The share of this sector in the gross domestic expenditure on R&D (GERD) has increased from 5% in 2013 to 7% in 2018.
 - The number of patent applications filed by the top 10 academic institutes and universities in India rose by more than two times in four years, from 838 in 2015-16 to 2,533 in 2019-20.

- **Key Problems in Patent Ecosystem:**

- **Long Pendency of processing Patent applications:**
- **Drastic increase in the number of abandoned patent applications:** The share of such abandoned patents application soared from 13.6% in 2010-11 to 48% in 2019-20.
 - Abandoned application refer to those applications where refiling of documents hasn't happened after patent examiner finds the original documents not enough to meet the requirements.
 - This may be due to the fact that applicants are not confident about their applications passing scrutiny and, therefore, do not pursue their applications.
 - Another reason may be the creation of perverse incentive for filing of Patents after the IPR Policy 2016. Innovators may have filed applications even when they knew that their claims may not pass scrutiny.
- **Decreasing Industry-Academia collaboration:** Though, our ranking in Global Innovation Index has improved over the last few years. But India's score on the indicator of Industry-Academia collaboration has declined and thus, India's rank on this indicator has also deteriorated.

- **Way Forward:**

- Increasing efficiency of processing patent applications.
 - Increased human resource
 - Fast tracked scrutinization process
- Examining the patent ecosystem to understand the reasons for large number of abandoned applications.
- **Increased Industry-Academia Collaboration:** Currently, it is limited to niche research areas that have low commercial significance.
- Other General steps:
 - Better implementation of IPR policy
 - Brining an innovation led ecosystem at schools, HEIs etc.
 - Prioritizing commercialization of the patents

- **Conclusion:**

- The Patent system is a critical aspect of the national innovation ecosystem, investing in the patent ecosystem will help in strengthening the innovation capability of India.
- Right interventions should be made for the promotion of the quality of patent application and collaboration between academia and industry.

5) GEOGRAPHICAL INDICATION

- **Example Questions**

- Discuss the key pre-requisites and advantages of getting Geographical Indication Status for any product. [15 marks, 250 words]

- **Introduction**

- A 'geographical indication' (GI) is a place name used to identify the origin and quality, reputation, or other characteristics of products. It is a sign used on products that have specific geographical origin and possess qualities or a reputation by virtue of their geographical association. The owner of the GI tag has exclusive rights over the product and can prohibit others from using the same name.

- For instance: champagne, Darjeeling tea, Nagpur Orange, Kangra Paintings etc.
- **India's GI Law** "Geographical Indications of Goods (Registration and Protection) Act, 1999" has come into force with effect from Sep 2003.
 - In India, the tag is awarded by the GI Registry in Chennai, and it indicates that a produce possesses certain qualities exclusive to its land of origin.
 - This GI registry administers the GI of Goods (Registration and Protection) Act, 1999, under the Controller General of Patents, Designs, & Trade Marks.
- **WTO Law:** GIs have been defined under Article 22(1) of the WTO Agreement on TRIPS.
- **Darjeeling tea** became the first GI tagged product in India, in 2004-05. Other famous GI products of India include Basmati Rice, Chanderi Fabric, Mysore Silk, Kanchipuram Silk, Banarasi Silk Saree, Jaipur Blue Pottery, Kullu Shawl, Kangra Tea, Thanjavur Painting, Allahabadi Surkha, Farrukhabad prints, Lucknawi Zardozi, Kashmir Walnut wood carving etc.
- **Significance of GI registration**
 - Legal protection -> Prevents unauthorized use of GI by others.
 - Consumer protection -> right information -> GI Tag conveys an assurance of quality and distinctiveness, which is essentially attributable to the place of its origin.
 - Promotes economic prosperity of the producers of goods by enhancing demand in national and international market
 - Essential to get protection in other countries.
 1. Article 22 of TRIPS agreement says unless a geographical indication is protected in the country of its origin, there is no obligation under the agreement for other countries to extend reciprocal protection.
 - **Opportunity to promote development in rural areas:** GI registration along with strengthening of e-commerce in rural areas can promote higher income for people producing GI tagged products.
 - The hyper-localized nature of GI offers solutions to reverse urban migration and conserve India's ancient crafts, culture and food.
 - **Other wider benefits** – Encourages protection of biodiversity, local know-how and natural resources.

Challenges and Way forward

- **GI businesses are micro;** therefore, it is necessary to address the challenges of capacity building, formal or easy access to credit, forming marketing linkages, research and development, product innovation and competitiveness, in both domestic and international market.
- **Solve the vexed issue of middlemen:** Ensure that distribution margins of intermediaries are competitive and doesn't exploit the producers.
- **Promote GI cooperative bodies or associations.** This can be nationally managed by a GI board under the auspices of the Department for the Promotion and Industry and Internal Trade (DPIIT).
- **Promote Digital literacy** among the GI producers. This should be a priority agenda for NGOs and other stakeholders like DPIIT

Conclusion: The Indian GI economy can be a platform for India to showcase to the world a model for ethical capitalism, social entrepreneurship, de-urbanization, and bringing women to workforce, on the back of robust digital system.

6. MISCELLANEOUS S&T TOPICS

1) TRANSFORMING S&T IN INDIA

- Example Questions:

- Science and Technology is the bedrock of economic growth, social progress and National Security. Discuss [12.5 marks, 200 words]
- Discuss the key factors hindering growth of Science and Technology in India. Suggest some way forward [15 marks, 250 words]

• Introduction: Why S&T is important

- Science, Technology and Innovation is rightly recognized as an engine of economic growth and PM Modi is right when he extended the old slogan of Lal Bahadur Shastri and Atal Bihari Vajpayee, "Jai Jawan, Jai Kisan, Jai Vigyan" to include "Jai Anusandhan (research and innovation)".
- They are not only key drivers of economic growth, but is also crucial for **social well-being** and **India's Security**.
 - Investing in science is also **fundamental to India's security**:
 - the human security of our population.
 - the resilience needed to deal with multiple challenges stemming from **climate change**;
 - and the national security challenges stemming from **new emerging threats** ranging from cyber warfare to autonomous military systems such as drones.
 - Growth in S&T also promotes **Scientific temper**, with its spirit of enquiry, the primacy accorded to facts and evidence, the ability to challenge status quo, the adherence to norms and discourse and the elevation of doubts and openness. This scientific temper **provides a bulwark against the darker forces of dogma, religious obscurantism and nativism** that are threateningly resurfacing around the world.

• Key Issues associated with S&T research and development in the country – Poor Input

- Stagnant Input in terms of investment in S&T:** Though the overall investment in S&T have shown a consistent increase over the last three decades but, as a fraction of GDP, public expenditure on research have been stagnant – around **0.6-0.7% of GDP**.
- Most of the Expenditure by Public Sector**
 - Public expenditure dominates the total S&T expenditure, though its share has come down from **3/4th to 3/5th (in last decade)**.
- Lack of contribution from Private Sector and state government:**
 - In most of the **other countries** private sector plays an important role in advancing R&D in science and technology. But in India, not only the government is the major investor, but is also the main user of this investment. According to Forbes in 2017, there were 26

Indian companies in the list of the top 2,500 global R&D spenders compared to 301 Chinese companies.

- Further the government expenditure on R&D is undertaken entirely by the central government without any contribution from state government

iv. **Relatively small role of universities in Research activities:**

- Globally, universities play a critical role in both creating the talent pool for research as well as generating high quality research output.
- In India, however, public funded research concentrates in **specialized research institutes**, under different government departments.
- Our universities play a largely a teaching role – a decision that goes back to 1950. It is now well recognized that whatever by the merit of the decision then, this disconnection has severely impaired both teaching as well as the research enterprise in the country.

v. **Less number of Ph.Ds. in Science, Technology, Engineering, and Mathematics (STEM)**

- Well trained workforce, especially the **Ph.D. students play an especially important role in promoting R&D in the country.**
- **Overall, though, India has far fewer researchers** than most R&D leaders in the world.

• **Poor Outputs in R&D in the country**

iii. **India produces fewer patents per capita:**

- According to WIPO, India is the 7th largest Patent Filing Office in the world.
- Still, India produces fewer patents per capita.

- **Expanding R&D in the country-Way Forward:** The above analysis clearly indicates that India needs to redouble its efforts to improve science and R&D in the country. Some of the following key measures can be taken.

i. **Doubling National Expenditure on R&D** should be the first and foremost step. Here, a lot of contribution needs to come from private sectors, universities and States

ii. **Focus on the foundation – Improve math and cognitive skills at the school level -> Quality of teaching; Freedom of thinking**

iii. **Encourage Investigator-led research**

- A step in this direction has been establishment of the **Science and Engineering Research Board (SERB)**, a statutory body of DST. The body has sanctioned more than 3,500 new R&D projects by individual scientists. This promising start needs to expand further.

iv. **Link National Lab and Universities**

- **The separation of research from teaching has been an Achilles heel for Indian Science.**
- **Current Scenario:** Universities have students but need additional faculty support, while research institutions have qualified faculty but are starved of bright young students brimming with energy and ideas.
- **How closer relation will help:** Students and researchers will complement each other. Together they can link up with the commercial sectors and help develop industrial clusters in those areas that draw on these research strengths and lay foundations of innovation driven “smart cities”.

v. **Take a Mission Driven Approach to R&D:** India has a potential to be global leader in a number of areas if it is willing to invest. However, it would require a deliberate focus in these areas.

Some missions which could be initiated includes National Mission on Dark matters, National Mission on Genomics, National Mission on Energy Storage Systems, National Mission on Mathematics, National Mission on Cyber Physical systems etc.

- vi. **Leverage Scientific Diaspora:** More than **1,00,000 Ph.D. holders** who were in India are now residing in different parts of the world.
 - vii. **Improve the culture of Research:** Less hierarchical governance system; greater representation of younger scientists in decision making bodies
 - viii. **Greater Public Engagement of Science and Research Establishment**
 - If science is to garner greater support from society, it will require scientists to engage more vigorously with society.
- **Conclusion:**
 - Only an improved focus on research and innovation will turn India's aspiration of becoming a Vishwaguru into reality and help it enter the golden era of Amrit Kaal.

2) DEEP OCEAN MISSION

- **Practice Questions:**
 - Discuss the key pillars of Deep Ocean Mission (DOM). Examine how the DOM will contribute towards India's sustainable economic growth [10 marks, 150 words]
- **Introduction:**
 - Deep Ocean Mission is India's quest to explore and harness the depths of the ocean.
 - One of its component is Samudryaan Mission which is aimed to develop **an indigenous, self propelled manned submersible** to carry **3 human beings** to a **water depth of 6,000 meters in the ocean** with a suite of scientific sensors and tools for deep ocean exploration.
- **The mission has six pillars:**
 - i. Development of technologies for deep sea mining and a manned submersible to carry three people to a depth of 6,000 meters in the ocean. The submersible will be equipped with suit of scientific sensors, tools, and an integrated system for mining polymetallic nodules from the central Indian Ocean;
 - ii. Development of ocean climate change advisory services
 - iii. Technological innovations for the exploration and conservation of deep-sea biodiversity.
 - iv. Surveys and exploration aimed at identifying potential sites of multi-metal hydrothermal sulphides mineralization along the Indian ocean mid-oceanic ridges.
 - v. Harnessing energy and freshwater from the ocean
 - vi. Establishing an advanced Marine Station for Ocean Biology, as a hub for nurturing talent and driving new opportunities in ocean biology and blue technology.
- **DOM** is one of the nine missions under the PM's Science Technology and Innovation Advisory Council (PMSTIAC). It is imperative that DOM supports the blue-economy priority area, blue trade, and blue manufacturing in India.
- **Progress:**

- **The National Institute of Ocean Technology (NIOT)**, an autonomous institute under MoES, has been entrusted with the mandate of developing indigenous technologies to address engineering challenges associated with exploring and utilizing ocean resources.

Matsya6000 is the deep ocean submersible designed to accommodate a crew of three members (aquanauts). It boasts of an operational endurance of 12 hours, which is extendable to 96 hours in the event of an emergency.

- **Progress:** The design of Matsya6000 has been completed.
 - Now tests will start, with a depth of 5,00 meters which will eventually be expanded for a depth of 6000 meters.



- NIOT has also successfully conducted deep-sea locomotion trials on the seabed at a depth of 5,270 meters using India's underwater mining system, 'Varaha'.
- **Why the target of 6000 meters depth?**
 - **Polymetallic modules**, which contain precious metals like copper, manganese, nickel, iron, and cobalt, are found approximately 5,000 m deep
 - **Polymetallic** sulphides occur at around 3,000 m in the central Indian Ocean.
- **Key challenges of India's DOM:**
 - **High Pressure in Deep Ocean:** Operating under such high pressure requires the use of meticulously designed equipment crafted from durable metals or materials. Electronics also find it difficult to function under this high pressure.
 - **Landing on the Ocean bed is another challenge.** It is due to incredibly soft and muddy sea bed.
 - **Extracting minerals** will also require a lot of energy as they will have to be pumped to the surface.
 - **The difficulty of propagation of EM waves** make it difficult to rove remotely operated vehicles.
 - **Visibility** poses another challenge as negligible light reaches a depth of 6,000 meters.
- **Where does India stand in comparison to other countries?**
 - Countries such as USA, Russia, China, Japan, and France have already achieved successful deep ocean crewed mission. India is poised to join the rank of these nations.
- **Way Forward:**
 - **Focus on Environment Sustainability:** The extraction of resources should be sustainable and should only minimally hamper the deep ocean ecosystem.
 - **International Collaboration** can be crucial in understanding the key challenges and their resolution from the countries which are already doing deep ocean exploration.
 - **Focus on training human resources** which would be specialized to work in the deep ocean systems.
- **Conclusion:**

- **DOM** with its focus on research and development, deep sea mining, biodiversity conservation and climate studies embodies the very essence of scientific zeal that India needs to move towards Atmanirbharta.

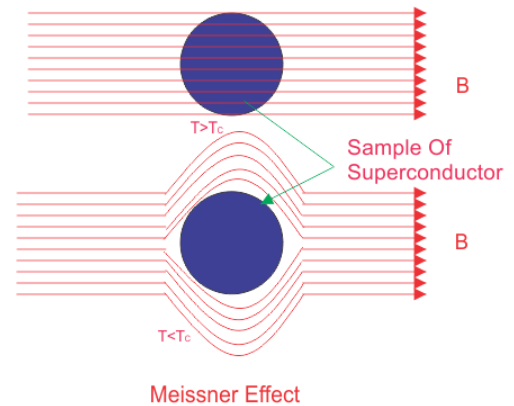
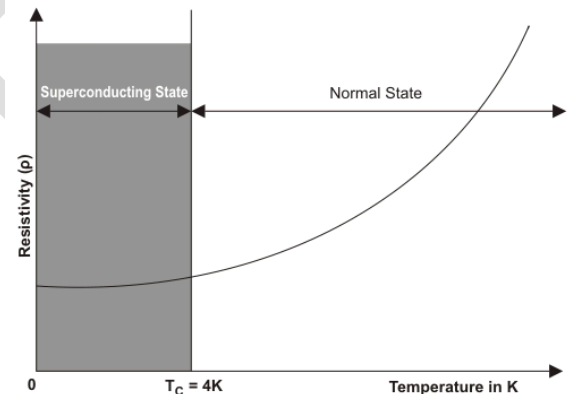
3) SUPERCONDUCTIVITY

Example Questions:

- What is superconductivity? Discuss its key applications. [10 marks, 150 words]

Definition

- Some materials when they are cooled below certain temperatures (T_{critical}), they lose all electrical resistivity. This is called superconductivity.
 - It is one of the nature's most intriguing quantum phenomenon and was first discovered more than 100 years ago in mercury cooled to temperature of liquid helium (-270 degree C) by Heike Kamerlingh-Onnes in 1911. He received 1913 physics Nobel Prize.
 - The temperature at which the metals change from normal conducting state to superconducting state is called **Critical/Transition** temperature.
 - For e.g. below 4 degree Kelvin the metal mercury becomes a superconductor, therefore critical temperature for mercury is 4 K.
 - The transition from normal conducting stage to superconducting stage is **reversible**.
- The super conducting material shows **some extra ordinary properties** which make them very important for modern technology. The research is still going on to understand and utilize these extraordinary properties of superconductors in various fields of technology.
 - **Infinite conductivity** (zero electric resistance)
 - **Persistent current**
 - **Meissner Effect**: a superconductor, expel the magnetic field and doesn't allow the magnetic field to penetrate inside it.
 - **Critical temperature**
 - **Critical magnetic field**
 - **Critical Current**
 - **Applications of Superconductivity**
 - **Medical Sector**: Used in magnetic resonance imaging, Magnetic Source imaging etc.
 - **Electric Engineering**: For generation of high performing generators, motors, transformers, relays, superconducting magnets etc.
 - **Electronics: Quantum Computing**, high quality sensors, filters, circuitry radar etc.
 - **Transportation**: Magnetically levitated trains, Marine propulsion motors etc.
 - **Fundamental Physics**: Particle accelerators, Magnets, Plasma / fusion research etc.



- **Superconductivity at Room Temperature???**

- The **holy grail of superconductivity** today is to **find or create materials that can transfer energy between each other in a non-pressurized container.**
 - If an efficient superconductor becomes possible at room temperature, it would revolutionize power transmission system for industry, commerce, and transportation.
- **Several Wrong Claims and Skepticism:** In recent years several claims of achieving superconductivity at room temperature has been found to be wrong. This has made scientific community a bit skeptic about any such new claim.
 - For e.g. in July 2023 only a research paper published in Physical Review Letters in 2021, by a US-based researcher making a similar claim had to be retracted.
 - Scientists at IISc Bengaluru had made similar claims in 2018, only to be sent for more reviews. The case is still unresolved.
- **In July 2023**, the South Korean researchers have posted two related papers on internet, not yet peer reviewed, claiming that a lead-based compound that they had developed had shown superconducting properties at room temperature, under normal pressure conditions. They are calling this material to be **LK-99**.

- **Conclusion**

- Superconductivity at room temperature and room pressure if achieved could revolutionize a number of sectors including energy, transportation, health etc. This remains a holy grail yet to be achieved

4) NOBEL PRIZE IN CHEMISTRY, 2023: QUANTUM DOTS

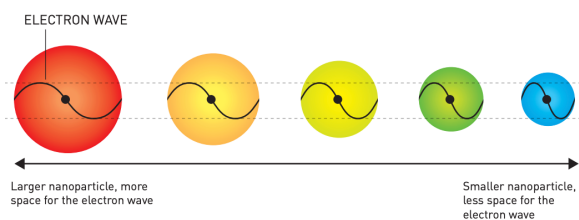
- **Quantum Dots** are nanoparticles so tiny that their size determines their properties.
- **Understanding Properties:** They have many fascinating and unusual properties. Importantly, they have different colors depending on their size.



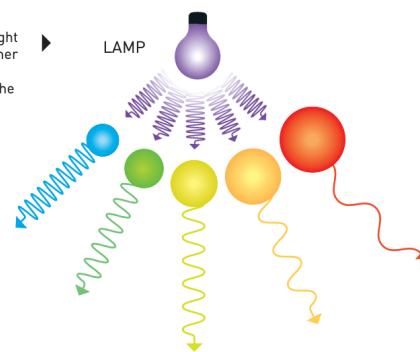
A quantum dot is a crystal that often consists of just a few thousand atoms. In terms of size, it has the same relationship to a football as a football has to the size of the Earth.

Quantum effects arise when particles shrink

When particles are just a few nanometres in diameter, the space available to electrons shrinks. This affects the particle's optical properties.



Quantum dots absorb light and then emit it at another wavelength. Its colour depends on the size of the particle.



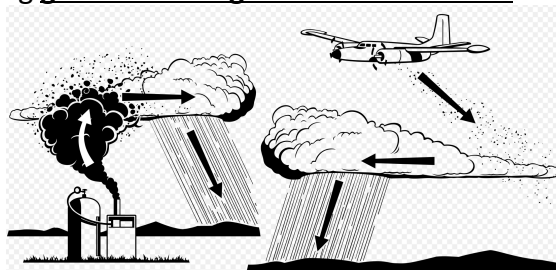
- **For decades**, Quantum phenomena in the nanoworld were just a prediction.
- **Contributions:**
 - **In the early 1980s**, Alexie Ekimov and Louis Brus succeeded in creating - independently of each other - quantum dots, which are nanoparticles so tiny that quantum effects determine their characteristics.

- **Alexie Ekimov**, in early 1980s, succeeded in creating size-dependent quantum effects in colored glasses.
 - The color came from nanoparticles of copper chloride and Ekimov demonstrated that the particle size affected the color of the glass via quantum effects.
 - This was the first time someone had succeeded in deliberately producing quantum dots - nanoparticles that cause size-dependent quantum effects.
- **Louis Brus**, a few years later, was the first scientist in the world to prove size-dependent quantum effects in particles floating freely in a fluid.
- **Moungi Bawendi**, in 1993, revolutionized the chemical production of quantum dots, resulting in almost perfect particles. This high quality was necessary for them to be utilized in applications.
- **Applications:**
 - Researchers have primarily utilized quantum dots to create colored light.
 - The luminous property of quantum dots are utilized in computer and television screens based on QLED technology, where the Q stands for quantum dot.
 - In these screens blue light is produced using the energy-efficient diodes that were recognized with the Nobel Prize in Physics 2014. **Quantum dots are used to change the color of some of the blue light, transforming it into red or green.** This makes it possible to produce three primary colors of light needed in a television screen.
 - **LED Lamps:** Quantum dots are used in LED lamps to adjust the cold light of the diodes. The light can then become as energizing as daylight or as calming as the warm glow from a dimmed bulb.
 - **Biochemistry and Biomedicine:** Biochemists attach quantum dots to biomolecules to map cell and organs. Doctors are also investigating the potential use of quantum dots to track tumour tissue in the body. Chemists instead use the catalytic properties of quantum dots to drive chemical reactions.
 - **Health Sector:** These can guide surgeons when they remove tumour tissues, among many other things.
 - **Future Applications:** Researchers believe that in the future they could contribute to flexible electronics, tiny sensors, thinner solar cells, and quantum cryptography.
- **Conclusion:**
 - Quantum Dots are bringing great benefits to humankind, and we have just begun to explore their potential.

5) CLOUD SEEDING

- **What is cloud seeding?**
 - It is a weather modification technique aimed at **enhancing precipitation from clouds.** The idea of cloud seeding was first conceived during WW-II and has since then become a much practiced activity in different dry regions of the world.
- **How does it work?**

- In cloud seeding, clouds are injected with salts like silver iodide, potassium iodide, or sodium chloride which act as seed. These salts provide additional nuclei around which more cloud droplets would form.
- There are **two principal cloud seeding techniques**:
 - i. **Hygroscopic Cloud Seeding**: It aims at speeding up droplet coalescence in liquid clouds, leading to production of large droplets that start to precipitate. Here seeding material is generally large salt particles.
 - ii. **Glaciogenic Cloud Seeding**: In this method the idea is to trigger ice production in supercooled clouds, leading to precipitation. It is usually done by dispersing efficient ice nuclei, such as silver iodide particles or dry ice (solid carbondioxide) into the cloud, causing heterogeneous ice nucleation.
- **How is cloud seeding done?**
 - It is done using **ground based generators or aircraft**.



- **What are the conditions required for cloud seeding to be done?**
 - **Moisture laden clouds**: Cloud seeding can only happen if there is sufficient cloud and sufficient depth of cloud.
 - **Wind speed below a certain level**
 - **Temperature** - cloud should be cold enough to contain supercooled liquid water
- **Application/advantages**
 - **Fighting water scarcity**: Rainfall in drought prone areas.
 - **Increasing winter snowfall** - which can supplement the natural water supply for communities in the surrounding area.
 - It can also be done to prevent hailstorm, dissipate fog etc.
 - **Increasing hydro power generation** (for e.g. in Tasmania, Australia)
 - **Fighting air pollution and Water pollution**
 - Rainfall can wash off pollution from air
 - More rainfall can also ensure ecological flow in rivers leading to reduced scope of pollution.
 - **Controlling forest fires**
 - **Atmospheric studies** - Studying cloud seeding can help scientists understand how normal cloud formation would occur.
- **Could cloud seeding be used to fight air pollution in Delhi?**
 - In India, **cloud seeding hasn't been tried for fighting pollution.**
 - **China** has tried this option.
 - In winters, cloud from over Delhi due to Western Disturbances and thus some experts suggested cloud seeding for rainfall to occur.

- **Has Cloud seeding been done before in India, and has it been successful?**
 - It has been attempted in Monsoon, in places such as Karnataka, Maharashtra, and Tamil Nadu.
 - A recent experiment, the fourth phase of the **Cloud Aerosol Interaction and Precipitation Enhancement Experiment (CAIPEEX-IV)** that took place in monsoon seasons of 2018 and 2019. It was conducted in drought-prone Solapur in Maharashtra. It pointed to **relative enhancement of 18% in rainfall**.
 - The approx. cost of producing water through this method was 18 paisa per litre, the cost will drop by more than 50% if we use indigenous seeding aircraft.

7. SOME ENVIRONMENT UPDATES

6) EB&CC: STATE OF WORLD FOREST REPORT: BY FOOD AND AGRICULTURE ORGANIZATION

- **Introduction:** The report provides a comprehensive analysis of forest conditions and trends.
- **Key Highlights:**
 - » **Forest Cover and Distribution:**
 - In 2020, the global forest covered approximately 4.1 billion hectares (or 31% of the land area).
 - **Russia, Brazil, Canada, USA and China** account for 54% of the world's forest in descending order.
 - » **Deforestation is slowing:** It fell from 15.8 million hectares per year between 1990 and 2002 to 10.2 million ha between 2015-2020. It reduced by 50% in Brazil's legal Amazon (between 2022-23).
 - » **10 countries recorded an annual gain in forest area in 2020:** It include China, Australia, India, Chile, Vietnam, Turkey, USA, France, Italy and Romania.
 - » **Climate change is making forest more vulnerable:**
 - **Wildfire intensity** is increasing.
 - **Insects and diseases are also emerging as threat:** For e.g. in the USA, 25 million hectares of forestlands are projected to experience losses exceeding 20% of host tree basal area due to insects and disease through 2027.
 - » **Demand for forest products is increasing:**
 - **Global wood production is at record level:** A total of 2.04 billion m3 of roundwood was harvested in 2022.
 - **Nearly 6 billion people use Non timber forest produce (NTFP)** including 2.77 billion rural users in global south.
 - **2050 projections** also indicate huge increase in wood demand (increase of 49% between 2020-2050)
- **Innovations is required to scale up Forest Conservation:**
 - » **Technological Innovation** - digital, product/process and biotechnological: For e.g., open access to remote sensing data and the facilitated use of cloud computing are enabling digital methodologies that generate forest data and improve forest management process.
 - » **Social innovation:** Better engage women, youth and indigenous people in developing locally led solutions.
 - » **Policy innovation:** Promotion of multi-stakeholder partnership and cross sectoral approaches in land use policies and planning.
 - » **Institutional innovation:** Support cooperatives to enhance the bargaining power of smallholders.
- **Practice Question:**
 - » **"The State of World Forest Report, 2024'** has brought both relief and concern regarding the state of world's forest" Elaborate [10 marks, 250 words]

7) EB&CC: INTERNATIONAL SOLAR ALLIANCE

- **Why in news?**
 - » Paraguay joins ISA as 100th full member (June 2024)
 - **Example Questions**
 - » What are the key objectives of International Solar Alliance? Discuss the key focus areas on which ISA is working to achieve these objectives. [12.5 marks, 200 words]
 - **Background**
 - » **Need of ISA:** Increasing significance of Solar energy in mitigating climate change and absence of any UN body specifically focused on Solar Energy.
 - » **Rationale behind ISA:** There are 121 countries which are located between Tropic of Cancer and Tropic of Capricorn. These countries have the potential to harness solar energy in a cost effective manner. This potential can be effectively realized if there are efforts towards experience sharing, technology sharing and finding locally appropriate solution.
 - **International Solar Alliance (ISA)**
 - » It is conceived as a coalition of solar resource rich countries and is aimed at trans regional solar energy cooperation. It
 - » It wants to accelerate the deployment of solar energy globally in support of climate action.
 - It was **jointly launched by India and France** on Nov 2015 on side lines of COP21 UNFCCC.
 - It is **headquartered** at campus of **National Institute of Solar Energy (NISE)** in Gurugram, India.
 - » **Goal:** Mobilize more than USD 1 trillion in investments to set up 1,000 GWs of solar installation globally, thereby making clean power affordable and universally accessible by 2030.
 - » **ISA Framework Agreement** entered into force on **6th Dec 2017**. It thus became a de-jure treaty based International Intergovernmental Organization, headquartered at Gurugram, India.
 - » **Membership:**
 - It has 119 signatory members of which **100 have ratified**.
 - In June 2024, Republic of Paraguay became the 100th full member of ISA.
 - **All UN members** are eligible to be members of ISA.
 - **Objectives**
 - » **Address key common challenges** to the scaling up of Solar energy in ISA member countries.
 - » **Mobilize investments needed** for massive deployment of Solar Energy.
 - Take **joint efforts to reduce the cost of finance** and the **cost of technology**.
 - Aims to **deploy more than 1,000 GW of solar energy** and mobilize over **\$1,000 billion** into solar power panel by 2030.
 - » Improve access to energy and opportunities for better livelihoods in rural and remote areas and to increase the standard of living.
 - » Identification of national opportunities to accelerate the development and deployment of existing clean solar energy technologies, the potential for which largely remains untapped.
- ISA is guided by 'Towards 1000' strategy:
- » Mobilize \$ 1,000 billion of investments in solar energy.

- » Deliver **energy access to 1000 million** people using clean energy solutions.
- » Installation of 1,000 GW of solar energy capacity.
- » Reduce carbon emissions to the tune of 1,000 million tonnes every year.
- **Other Significance**
 - » Key step in increasing collaboration to fight **environmental pollution and climate change.**
 - » Example of India's Progressive and Cooperative Climate Engagement. Formation of ISA underlined India's presence as a dominant global force in the challenging politics of climate change.
 - » **Geopolitical:** Transregional solar energy cooperation can pave the way for equability and a just energy order in the International arena.
- **Some Problems which ISA is facing**
 - » **Coordination issues:** Issues like membership rights; procedure over pragmatism are creating hinderance.
 - » **China's dominance** in the solar supply chain is another hindrance.
 - » **Issues of land acquisition and environmental clearance**
 - » **Grid Integration** is another major technical issue.
 - » **Financing and regulatory hurdles** continue to hamper the growth of solar sector.
 - » **Lack of public Understanding about ISA's role:** Despite the ball moving behind the scenes, public understanding about ISA's role, scope and future direction remains limited.
- **Way forward:**
 - » Increased cooperation.
 - » Diversification of manufacturing capacity among ISA countries.
 - » Innovative solutions to reduce land requirements
 - » Fast tracking research on effective grid integration.
 - » Phased targets for mobilization of finance.

8) EB&CC: WETLANDS AND THEIR SIGNIFICANCE

- **Why in news?**
 - » Three more Ramsar wetland sites now in India, taking the total to 85 (Aug 2024)
- **Practice Questions:**
 - » What is wetland? Explain the Ramsar concept of 'wise use' in the context of wetland conservation. Cite two example of Ramsar sites from India. [150 words, 10 marks] [Mains 2018]
 - » Discuss the major ecological functions and services provided by wetlands. Enumerate the key challenges faced by wetlands in India.
- **What is a Wetland?**
 - » **A Wetland** is a **transitional land between terrestrial and aquatic ecosystem.** It is an ecosystem that is flooded either permanently or seasonally.
 - **UN Ramsar Convention** defines wetlands as 'areas of marsh, fen, peatland or water, weather natural or artificial, permanent or temporary, with water that is static or

*flowing, fresh brackish or salt, including areas of marine water the **depth of which at low tide doesn't exceed six meters**'.*

- **Significance of Wetlands:** Wetlands provide a wide range of important resources and ecosystem services such as:
 - » **Water and Food Security:**
 - Rice cultivation, fishery etc.
 - » **Ecosystem Services:**
 - Ground water recharge; Water Purification (e.g. East Kolkata Wetlands)
 - **Biodiversity** -> wetlands are transition zones between two different ecosystems and thus are highly productive.
 - » **Disaster mitigation: Buffer** from floods, droughts, cyclones and other extreme events.
 - For e.g. Deepor Beel of Guwhati serves as the flood prone city's natural stormwater drain.
 - » **Climate Change Mitigation and Adaptation Action:** Wetlands are important carbon sink and facilities for rainwater storage.
 - » **Tourism** is another area where Wetlands such as large lakes have played crucial role.
 - For e.g. Ashtamudi lake of Kerala is one of the most popular tourist attractions in Kollam.
 - » **Coastal Protection** is ensured by Wetlands as they act as buffer zones.
- **Threats faced by Wetlands of India:** As per the recently released IPBES assessment, wetlands are the most threatened ecosystem. This impacts **40% of the world's plants and animals** that live or breed on wetlands.
 - **Water Pollution: Disposal of untreated sewage; Dumping of municipal waste** etc are threatening the natural functioning of the wetlands.
 - **Noise Pollution:** It is caused by rapid urbanization around the wetlands and continues to be an irritant and is putting migratory visitors away.
 - **Encroachment of wetlands** in the form of haphazard real-estate development.
 - **Proliferation of invasive species** such as water hyacinth, salvinia etc.
 - **Very less focus on** small non-notified wetlands.
 - Absence of any inventory related to wetlands.
 - **Lack of resources with local bodies** to ensure proper care and protection of the wetlands.
 - **Unclear and overlapping jurisdictions** between various government agencies (for example - wetlands in Delhi are under the jurisdiction of Delhi Jal Board, Delhi Urban Shelter Improvement Board, Public Works Department, ASI and Forest Department.
 - But other than Delhi Jal Board and Forest Department, none of the other bodies have any expertise in management of wetlands.
- **Key Efforts for Wetland Conservation:**
 - **International Efforts**
 1. **Ramsar Convention**
 - **Introduction**
 - The Convention on Wetlands of International Importance, called the Ramsar Convention, is the intergovernmental treaty that provides the

framework for the conservation and wise use of wetlands and their resources.

- **The aim** of the Ramsar list is "to **develop and maintain an international network of wetlands** which are important for the conservation of global biological diversity and for sustaining human life through the maintenance of their ecosystem components, processes and benefits".
- **Concept of 'Wise use'** is at the centre of Ramsar Convention.
 - Through this, the convention continues to emphasize that **human use on sustainable basis is entirely compatible** with Ramsar principles and wetland conservation in general. **Application of "wise use" concept is crucial to ensure that wetlands continue to support biological diversity as well as human well-being.**
 - The wise use guidelines emphasize on:
 1. **Adoption of national wetland policies**, involving review of local legislation and institutional arrangements to deal with wetland matters.
 2. **Development of programs** of wetland inventory, monitoring, research, training, education etc.
 3. **Take action at wetland sites**, involving the development of integrated management plans covering every aspect of the wetlands and their relationships.
 - The concept applies to all wetlands and water resources in contracting parties territories (not just to Wetlands of International Importance)
- **India** is a party to the Convention since 1982 and has so far declared 85 wetlands as Ramsar sites covering 23 states/UT.

- National Efforts

1. **Mapping of Wetlands:** As per the "National Wetland Decadal Change Atlas, 2017 published by Space Application Centre - ISRO Ahmedabad, a total of 2,31, 195 wetlands (are ≥ 2.25 ha) have been mapped in the country.
2. **Wetland (Conservation and Management) Rules 2017**, was notified by MoEF&CC under the EPA, 1986. It replaced the 2010 rules.
 - **Decentralization** -> empowers states and Uts to identify and manage their wetlands.
 - **Institutions:**
 - State Wetland authorities (SWAs) headed by State environment minister, established in each state and UT
 - The SWAs are responsible for effective conservation, management and monitoring of ecological condition of wetlands within their jurisdiction.
 - **National Wetland Committee to replace Central wetland regulatory authority and is responsible for monitoring the implementation of these rules**
 - **Banned activities like dumping solid waste, electronic etc.**
 - **Applicability**

- Wetland classified under RAMSAR
 - Wetland notified by Central, state or UTs.
3. **Comprehensive Guidelines** for the implementation of the above rules have also been published by MoEF&CC
 -
 4. **National Plan for Conservation of Aquatic Ecosystem (NPCA)** by MoEF&CC
 - It is a single conservation program for both wetlands and lakes and was formulated in 2015 by merging National Lake Conservation Program and National Wetland Conservation Program.
 - It covers various activities like interception, diversion, and treatment of waste water; shoreline protection; lake front development etc.
 5. **Mission Sahbhagita**: MoEF&CC has also launched Mission Sahbhagita, on the occasion of Azadi Ka Amrit Mahotsav, which is important step towards participatory conservation and wise use of wetland to enable a society ownership approach with communities leading at the forefront.
 6. **Save Wetland Campaign** has also been launched by the Ministry as part of the Mission Sahbhagita.

- **Way forward for wetland Protection in India**

- i. **Identification / Inventorization** of all the small/big wetlands in the country
- ii. **Implementation of various rules** regarding protection of wetlands, prohibition of dumping, pollution, Sewage treatment etc.
- iii. **Restoration of degraded wetlands**
- iv. **Strengthen monitoring**: Empower ULBs and PRI institutions for the purpose.
- v. **Involve Local People in wetland conservation**: Especially farmers, and fishing community who are depending on wetlands for their livelihood.
- vi. **Integrating Wetlands in Urban Planning**:
 - At urban level, wetlands management should be assigned to local municipalities to increase public accountability.
 - At the same time, there is a need to strengthen municipalities through proper allocation of resources.

- **Conclusion:**

- » Coherent management and governance of wetlands in India will not only secure ecological balance, but will also provide protection from climate risks and add to economic value of the region through employment generation in multiple forms.